

QXU5032 - Polymer Physics Presentation Assessment (20% of overall marks)

The Polymer Physics Presentation Assessment will evaluate your understanding of key concepts and topics in polymer physics, your ability to effectively communicate these concepts, and your presentation skills.

For the topic of the presentation, you should carefully consider your interests, background knowledge, and the course objectives. It's essential to choose a topic that genuinely intrigues you, as your enthusiasm will shine through in your presentation. Reviewing course materials, exploring recent research trends, and identifying practical applications can help narrow down your focus. Striking a balance between a topic that is challenging yet manageable within the given timeframe is key. Ultimately, selecting a topic aligned with your interests and course objectives will lead to a more engaging and successful presentation. Some example topics are listed at the end of this document but you could certainly select a different one. **You CAN select the same topic as the one of your group essay.**

This assessment element is intended to assess both the content and delivery of the presentation. The PowerPoint slides of the group presentations should be uploaded at QMPLUS one day before the actual presentations start taking place (**Deadline: 14/11 – Presentations to take place 17-18/11**). You will strictly have 7 minutes at maximum to deliver your presentation and at least 2 group members should deliver the contents (if more group members want to present, this is welcome). A single mark will be allocated to all group members.

Assessment Criteria:

Content Knowledge (40 marks):

Understanding of Polymer Physics Concepts: depth of your understanding of fundamental polymer physics concepts, such as polymer structure, behavior, and properties.

Clarity of Content: organization and clarity of the presentation, ensuring that it effectively conveys the key concepts.

Relevance and Significance: ability to highlight the relevance and importance of polymer physics in real-world applications.

Incorporation of Visual Aids: use of appropriate visual aids (e.g., diagrams, graphs, images) to enhance understanding.

Presentation Skills (30 marks):

Delivery: speaker's confidence, pace, and overall delivery of the presentation.

Engagement: presenter's ability to engage the audience through effective communication, eye contact, and body language.

Time Management: presenter's ability to adhere to the allotted presentation time.

Q&A Handling: presenter's ability to respond to questions from the audience with clarity and confidence.

Visual Design (15 marks):

Slide Design: the visual design of presentation slides, including clarity, readability, and alignment with the content.

Visual Consistency: the consistency of visual elements, such as fonts, colors, and formatting, throughout the presentation.

Overall Impressions (15 marks):

Overall Impact: the overall impact and effectiveness of the presentation.

Originality and Creativity: the level of originality and creativity in the presentation approach.

Example Topics:

- Polymer Classification: Understanding the Different Types.
- Copolymerization: Blending Different Polymers.
- Molecular Weight and its Significance in Polymers.
- Polymer Processing Methods: Molding, Extrusion, and More.
- The Glass Transition Temperature in Polymers.
- Polymer Crystallinity: Structure and Properties.
- Viscoelasticity of Polymers: Behavior and Applications.
- Polymer Melts: Flow and Rheology.
- Polymer Characterization Techniques: Spectroscopy, Chromatography, and More.
- Polymer Blends and Alloys: Properties and Applications.
- Biopolymers: Nature's Polymers and Their Applications.
- Synthetic Polymers in Medicine: Biomaterials and Drug Delivery.
- Polymers in the Food Industry: Packaging and Preservation.
- Polymers in Electronics: Conductive Polymers and Applications.
- Polymer Nanocomposites: Enhancing Material Properties.
- Sustainable Polymers: Biodegradable and Eco-Friendly Materials.
- Polymers in 3D/4D Printing: Innovations and Challenges.
- Polymers in Coatings and Paints: Durability and Performance.
- Polymer Nanoparticles: Applications in Medicine and Industry.
- Polymers in Adhesives and Sealants: Bonding Solutions.
- Polymer Foams: Structure, Properties, and Uses.
- Polymers in Textiles: Comfort, Durability, and Innovation.
- Smart Polymers: Responsive Materials and Applications.
- Polymers in Water Treatment: Purification and Filtration.
- Polymers in Renewable Energy: Solar Cells and Batteries.
- Polymer Recycling: Challenges and Sustainability.
- Polymers in Aerospace: Lightweight and High-Performance Materials.
- Polymers in Automotive Engineering: Safety and Efficiency.
- Conducting Polymers: Applications in Electronics and Sensors.
- Biodegradable Polymers: Reducing Environmental Impact.

- Polymer Gels: Properties and Applications.
- Polymers in Cosmetics: Formulations and Innovations.
- Polymers in Sports Equipment: Performance and Safety.
- Antimicrobial Polymers: Hygiene and Healthcare.
- Polymers in Robotics: Flexible and Soft Robotics.
- Polymer Sensors: Applications in Industry and Medicine.
- Polymers in Art and Design: Creative Applications.